

WASP-1b

R-1615

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_160811 · created 2026-08-02T03:49:34+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457611.8099 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.031
Transit depth (Rp/Rs)^2	1.94 +/- 0.86 [%]
Orbital Inclination (inc)	89.65 +/- 1.44
Airmass coefficient 1 (a1)	0.966 +/- 0.038
Airmass coefficient 2 (a2)	0.025 +/- 0.035
Scatter in the residuals of the lightcurve fit is	3.89 %
Best Comparison Star	#2 - [233, 56]
Optimal Method	PSF photometry
Transit Duration (day)	0.1104 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.139 ± 0.031 vs 0.1036 (deviation 0.93σ)
Duration fit vs published	0.1104 d vs 0.1567 d (deviation 4.96σ)
Mid-time O–C	19.4 min
Depth SNR	3.7
Residual scatter	3.89 %

Light curve

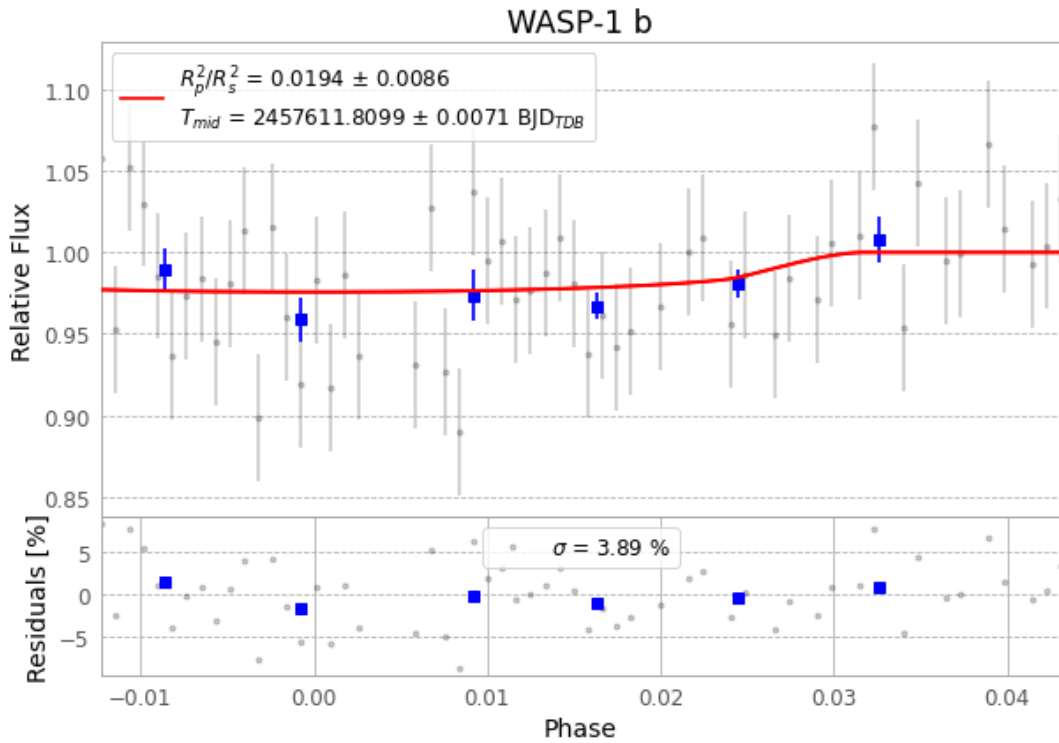


Figure 1 — FinalLightCurve_WASP-1 b_2016-08-10.png (SHA-256 1acaa0b0af6657bf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [24, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6e3d34e9a9e2dff1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

58 FITS frames (38 MB), WASP-1160811063612.FITS → WASP-1160811095712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-160811.log (391f37646f89...), FinalLightCurve_WASP-1 b_2016-08-10.png (1acaa0b0af66...), FinalLightCurve_WASP-1 b_2016-08-10.csv (64b9223a3448...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 03:49Z.